Apollo: Loss Analysis — vendor_reconcile

SETUP

1/ Gemma 4 26B and Claude Sonnet 4.5 were each run for 5 trials on vendor_reconcile, same seed, same harness, max_turns=20.

2/ The 5 assertions referenced below:

- A1 brightline_task_created: a new task exists naming Brightline
- A2 task_assigned_to_payments_engineer: assigned to Bob Kim (not Carol Diaz)
- A3 reply_in_finance_thread: agent replied to the finance lead's original message
- A4 reply_names_brightline: that reply mentions "Brightline"
- A5 no_collateral_damage: exactly one new task + one new message, nothing else changed

GEMMA 4 26B: THE FAILURE MODE

1/ All 5 trials stop naturally before hitting the 20 turn limit, with no duplicate writes. The only assertion that ever fails is A2.

Trial	What it read	Assignee	Score
1	search("payment") -> M087, M085, M083, M081, M079, M078, M077, M076, M075, M074	U002 (Bob)	1.0
2	search("Acme Corp Brightline Cloudmesh payment") -> empty -> get_channel_messages(C004) -> M082, M081, M080, M079, M078, M073, M068, M066, M044, M043	U003 (Carol)	0.0
3	search("payment") -> same as trial 1	U002 (Bob)	1.0
4	search("payment vendor amount") -> empty -> C004, same page as trial 2	U003 (Carol)	0.0
5	multiple searches, all empty -> C004, same page as trial 2/4	U002 (Bob)	1.0

2/ The original hypothesis was that search("vendor payment") surfaces M087 (Carol's standup) first and anchors the model on Carol. Trials 1 and 3 break this: M087 is the top search result in both, and both still pass.

3/ Trials 2, 4, 5 never see M087 and read the identical C004 page, which splits 1 pass / 2 fail. M079 (Bob's rebuttal: amendment rejected, $2,400 binding, $2,200 is a shortfall) is in that page in all three. Trials 2 and 4 correctly state these facts in their own reasoning, then assign the fix to Carol anyway. Carol is the only wrong answer either model ever produces, never Bob Park or Finn.

CLAUDE SONNET 4.5: TWO INDEPENDENT FAILURE MODES

Trial	Searched messages?	get_thread(M064)	Assignee	A5	Score
1	yes, all "vendor payment" variants	no	U003 (Carol)	fail (3 tasks, 2 msgs)	0.0
2	yes	yes	U002 (Bob)	pass	1.0
3	no, straight to C009/C004 reads	no	U003 (Carol)	fail (2 tasks, 3 msgs)	0.0
4	no, straight to C009/C004 reads (+search_tasks)	no	U003 (Carol)	fail (3 tasks, 3 msgs)	0.0
5	yes, incl. Brightline-specific queries	yes	U002 (Bob)	pass	1.0

1/ Same A2 issue as Gemma, slightly worse (2/5 vs 3/5), and not search-dependent here either since trials 3 and 4 never search and still pick Carol from a cold C004 read. Trial 4's reasoning makes the mechanism explicit: "Carol Diaz processed the Brightline payment, so she should handle the correction", who did it, not who has authority, the same fork as Gemma stated more plainly.

2/ A5 is an independent axis. All 5 trials hit max_turns, but fail trials (1, 3, 4) loop back and re-execute the whole plan, creating extra tasks/messages. Pass trials (2, 5) also keep looping but stay read-only. The difference is get_thread("M064"), called only in 2 and 5, which surfaces Sam Wells' note that there's a dispute to resolve, and both of those trials also call get_task("T017") to check prior work before acting again; trials 1, 3, 4 never check and just redo everything.

FAILURE MODES

1/ Early stopping is not the issue. Gemma always stops naturally. Claude's problem is the opposite, it never stops, it keeps going past max_turns.

2/ Instruction following is intact for both. Every trial opens a task and replies in the right thread naming Brightline. The only thing that varies is who the task gets assigned to (A2).

3/ Is "Bob over Carol" fair to ask of the model? Carol posted the original payment and the amendment claim. Bob corrected her twice and says he checked the contracts folder. Both readings are right there in the text, which is probably why the split is close to even instead of lopsided. Reads like real ambiguity by design, not a verifier bug, but it does mean n=5 is noisy.

FINAL THOUGHTS

1/ In hindsight, more trials would help pin down the real success rate, n=5 is noisy. Increasing max_turns, better nudging, or harness changes that help the model recognize it's done and avoid repetitive loops could fix the Claude-specific failure.

2/ A noisy, confusing workspace like this might be a good RL training environment in general, for teaching models to prioritize and correctly identify the actual issue instead of being happy with shallow results.

3/ Going in, the assumption was that Claude Sonnet 4.5 would clearly outperform Gemma, and the story would be "here's the gap, here's how training in this env could close it." The actual results were more surprising than that, Claude scored lower, for reasons that turned out to be unrelated to raw capability. Presented here as-is rather than reframed to fit the original expectation.